

2008



Mathematics

Extension 2

General Instructions

- Working time: 1½ hours
- Reading time: 5 minutes
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.

Total marks – 75

- Attempt Questions 1 – 3
- All questions are of equal value
- Begin each question in a new booklet
- All necessary working should be shown in every question.

Question 1 – (25 marks) – (Start a new booklet)

Marks

a) Consider the Hyperbola $H: \frac{x^2}{4} - \frac{y^2}{9} = 1$

- Find the x -intercepts
- Find the eccentricity e
- Find the foci
- Find the directrices
- Find the asymptotes
- Sketch H clearly showing all of the above features

1
1
1
1
1
2

b) Consider the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- Prove that the equation of the tangent to E at $P(x_1, y_1)$ is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
- The tangent to E at the point P cuts a directrix at the point T . Prove that the line segment PT subtends an angle of 90 degrees at the corresponding focus.
- Using the basic definition of an ellipse prove that $PS + PS^1 = 2a$ where S, S^1 are the two foci of E .

3
4
2

c) It can be shown that the line $y = mx + k$ is tangential to the hyperbola $H: \frac{x^2}{4} - \frac{y^2}{15} = 1$ if $4m^2 - 15 = k^2$

- Find the equation of the line through $(-1, 3)$ with gradient m
- Hence find the equations of the two tangents to H from $(-1, 3)$

1
4

d) The equation of the chord of contact from an external point $P(x_0, y_0)$ to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$.

- State the condition which x_0 and y_0 must satisfy if P is external to the ellipse.
- Find the condition which c must satisfy if $x + y = c$ is a chord of contact for the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

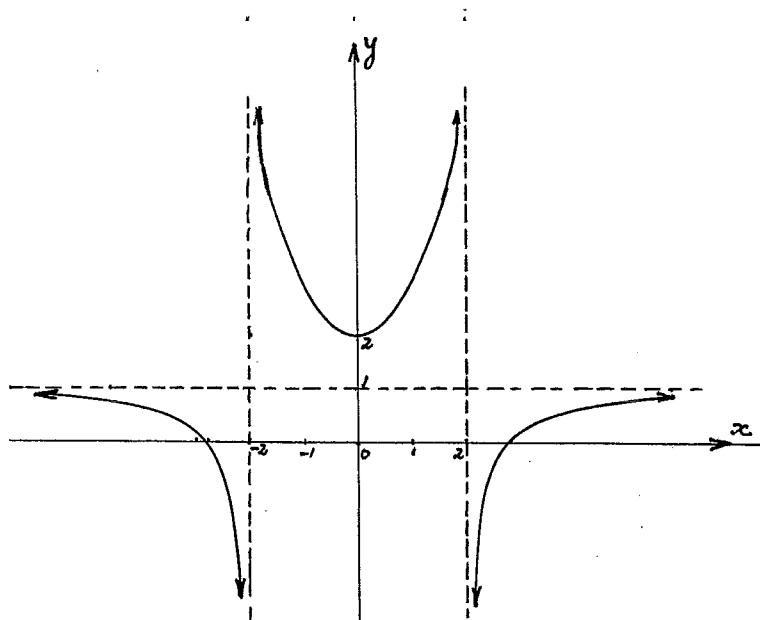
1
3

Question 2 – (25 marks) – (Start a new booklet)

Marks

- a) The graph of $y = f(x)$ is shown below.

On the separate grids provided, sketch



- | | |
|--------------------------|---|
| (i) $y = \frac{1}{f(x)}$ | 2 |
| (ii) $y = \sqrt{f(x)}$ | 2 |
| (iii) $y = [f(x)]^2$ | 2 |
| (iv) $y = \log [f(x)]$ | 2 |
| (v) $y = x + f(x)$ | 2 |
| (vi) $y = f^{-1}(x)$ | 2 |

Question 2 (cont'd)

Marks

- b) We aim to sketch the function $\sqrt{x} + \sqrt{y} = 1$

- | | |
|--|---|
| (i) Determine the x and y intercepts made by this function | 1 |
| (ii) Determine the domain and range of this function | 2 |
| (iii) Show that this function is decreasing for all x in the domain | 2 |
| (iv) Determine any stationary points and critical points | 3 |
| (v) Find $\frac{d^2y}{dx^2}$ and comment on its significance in the domain | 3 |
| (vi) Sketch the function | 2 |

Question 3 – (25 marks) – (Start a new booklet)

Marks

- a) Consider the function $f(x) = \frac{e^x - 1}{e^x + 1}$
- (i) Find the x intercept 1
 - (ii) Determine the behaviour of $f(x)$ as $|x|$ becomes very large (ie x approaches positive or negative infinity) 2
 - (iii) Show that $f(x)$ is an increasing function 2
 - (iv) Sketch the function $y = f(x)$ 2
 - (v) Discuss the number of solutions of $f(x) = mx$ for varying values of m . 3
- b) Consider the rectangular hyperbola : $xy = 4$. $P(2p, \frac{2}{p})$ and $Q(2q, \frac{2}{q})$ are variable points on R with $p > 0, q > 0$. M is the mid-point of the chord PQ .
- (i) State the foci of R 1
 - (ii) State the equations of the two directrices of R 1
 - (iii) Find the gradient of the chord PQ 2
 - (iv) Find the equation of the chord PQ 2
 - (v) If P and Q vary such that the line PQ always passes through the point $(4, 0)$ find the locus of M in the form $x = a$ 3
 - (vi) Find the range of this locus 1
- c) The point $P(a \sec \theta, b \tan \theta)$ lies on $H: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The normal at P cuts the x axis at A and the y axis at B .
- (i) Prove that the normal to H at the point P has equation $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$ 3
 - (ii) Prove that $PA:PB = b^2:a^2$ 2

Extension 2. Solutions

Q1
a)(i) $\frac{x^2}{4} - \frac{y^2}{9} = 1$

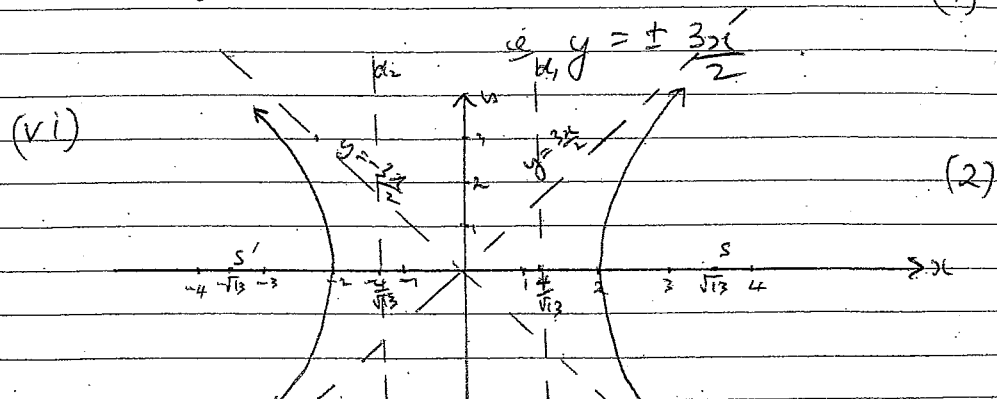
$y=0 \Rightarrow x = \pm 2$. $\therefore (2,0), (-2,0)$ are x -intercepts. (1)

(ii) $b^2 = a^2(e^2 - 1)$ $a=2, b=3$.
 $\therefore 9 = 4(e^2 - 1)$
 $\therefore e^2 - 1 = \frac{9}{4}$
 $\therefore e^2 = \frac{13}{4}$
 $\therefore e = \frac{\sqrt{13}}{2}$ (1)

(iii) Foci are $(\pm ae, 0)$ (1)
 $\therefore (\pm \sqrt{13}, 0)$

(iv) Directrices are $x = \pm \frac{a}{e}$ (1)
 $\therefore x = \pm \frac{4}{\sqrt{13}}$

(v) Asymptotes are $y = \pm \frac{bx}{a}$ (1)



b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(i) $\frac{d}{dx} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = 0$

$\therefore \frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$

$2yy' = -\frac{2x}{a^2} \times b^2$

$\therefore y' = -\frac{b^2}{a^2} \frac{x}{y}$

at $P(x_1, y_1)$, $y' = -\frac{b^2}{a^2} \frac{x_1}{y_1}$

eqn of tangent is
 $y - y_1 = -\frac{b^2}{a^2} \frac{x_1}{y_1} (x - x_1)$

$\times$ both sides by $\frac{y_1}{b^2}$

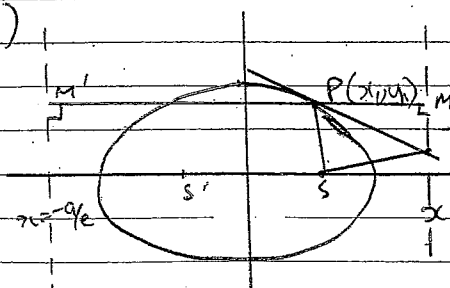
$\Rightarrow \frac{yy_1}{b^2} - \frac{y_1^2}{b^2} = -\frac{x_1x}{a^2} + \frac{x_1^2}{a^2}$

(3)

$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2}$

$= 1$ since (x_1, y_1) lies on ellipse

(ii)



at T, $x = a/e$

$\therefore \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ because

$\frac{a}{ea^2} x_1 + \frac{y_1}{b^2} = 1$

$\therefore \frac{x_1}{ae} + \frac{y_1 y_1}{b^2} = 1$

$$\frac{yy_1}{b^2} = 1 - \frac{x_1}{ae}$$

$$y = \frac{b^2}{y_1} \left(1 - \frac{x_1}{ae} \right)$$

$$\therefore T \left(\frac{a}{e}, \frac{b^2}{y_1} \left(1 - \frac{x_1}{ae} \right) \right)$$

$$\text{Now, slope PS} = \frac{y_1}{x_1 - ae} = m_1$$

$$\text{slope TS} = \frac{\frac{b^2}{y_1} \left(1 - \frac{x_1}{ae} \right)}{\frac{a}{e} - ae} = m_2$$

$$m_1 m_2 = \frac{y_1}{x_1 - ae} \times \frac{b^2 \left(1 - \frac{x_1}{ae} \right)}{y_1 (a/e - ae)} \times \frac{ae}{ae}$$

$$= \frac{y_1}{x_1 - ae} \times \frac{b^2 (ae - x_1)}{y_1 (a^2 - a^2 e^2)}$$

$$= \frac{-b^2}{a^2(1-e^2)}$$

(3)

$$= -1 \quad \text{since } b^2 = a^2(1-e^2) \text{ for ellipse.}$$

$$\therefore PS \perp TS$$

$\hookrightarrow$ PT subtends 90° at focus.

(ii) $PS = e PM$, M is foot of perpendicular to corresponding directrix.

and similarly, $PS' = e PM'$

$$(2) \quad PS + PS' = e (PM + PM')$$

$$= e (MM')$$

$$= e \times 2a$$

$$= 2a$$

$$c) (i) \quad y - 3 = m(x + 1) \quad (1)$$

$$\hookrightarrow y = mx + m + 3 \quad \text{or } K = m + 3$$

(ii) For tangents to exist,

$$4m^2 - 15 = K^2$$

$$= (m + 3)^2$$

$$\therefore 4m^2 - 15 = m^2 + 6m + 9$$

$$\therefore 3m^2 - 6m - 24 = 0$$

$$m^2 - 2m - 8 = 0$$

$$(m - 4)(m + 2) > 0$$

(4)

$$\therefore m = -2, 4$$

$\therefore$ tangents are

$$y = -2x + 1, \quad y = 4x + 7 \text{ using } *$$

$$d) (i) \quad \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} > 1 \quad (1)$$

$$(ii) \quad x + y = c \text{ corresponds to } \frac{x x_0}{a} + \frac{y y_0}{b}$$

$$\text{or } \frac{x}{c} + \frac{y}{c} = 1$$

$$\Rightarrow \frac{x_0}{a} = \frac{1}{c}$$

$$\frac{y_0}{b} = \frac{1}{c}$$

$$\therefore x_0 = a/c, \quad y_0 = b/c$$

Now, we require $\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} > 1$ from (1)

$$\therefore \left(\frac{a}{c} \right)^2 + \left(\frac{b}{c} \right)^2 > 1$$

$$\frac{a^2}{c^2} + \frac{b^2}{c^2} > 1$$

$$\therefore a^2 + b^2 > c^2$$

$$\frac{13}{c^2} > 1$$

$$c^2 < 13$$

$$\therefore \sqrt{13} < c < \sqrt{13}$$

(3)

Q2 a) see separate sheets.

Q2 b) (i) x intercept (1, 0) (1)

y " (0, 1)

(ii) D: $0 \leq x \leq 1$ (2)

R: $0 \leq y \leq 1$

(iii) $\sqrt{x} + \sqrt{y} = 1$
 $\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{-\frac{1}{2}} y' = 0$ (2)

$$\therefore y' = -\frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}} = -\frac{\sqrt{y}}{\sqrt{x}} < 0 \quad \forall x \in \mathbb{R}$$

(iv) $y' = 0$ at (1, 0) (3)
 y' undef'd at (0, 1)

(v) $y' = -\frac{\sqrt{y}}{\sqrt{x}}$

$$\therefore y'' = \sqrt{x} \cdot -\frac{1}{2}y^{-\frac{1}{2}} y' + \sqrt{y} \cdot \frac{1}{2}x^{-\frac{3}{2}}$$

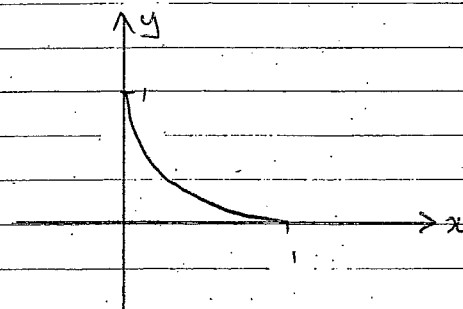
$$= \frac{-\sqrt{x}}{2\sqrt{y}} \cdot -\frac{\sqrt{y}}{\sqrt{x}} + \sqrt{y} \cdot \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{2} + \frac{\sqrt{y}}{2\sqrt{x}}$$

$$= \frac{\sqrt{x} + \sqrt{y}}{2\sqrt{x}} > 0 \text{ for } \forall x \text{ in domain}$$

curve is concave up

(vi)



(2)

Q3

a) $f(x) = \frac{e^x - 1}{e^x + 1}$

(i) $\frac{e^x - 1}{e^x + 1} = 0 \Rightarrow x = 0$ at (0, 0) (1)

(ii) as $x \rightarrow \infty$, $\frac{e^x - 1}{e^x + 1} \rightarrow 1$ (2)
as $x \rightarrow -\infty$, $\frac{e^x - 1}{e^x + 1} \rightarrow -1 = -1$

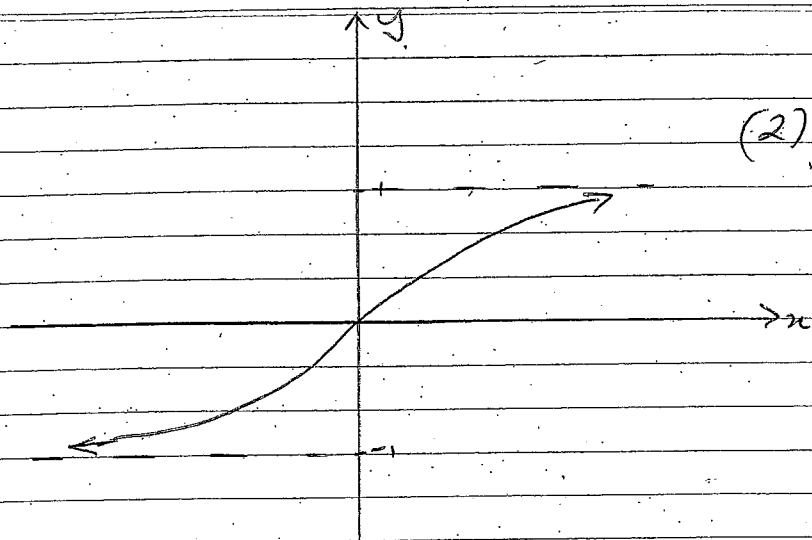
(iii) $f'(x) = \frac{(e^x + 1)e^x - (e^x - 1)e^x}{(e^x + 1)^2}$
 $= \frac{e^{2x} + e^x - e^{2x} + e^x}{(e^x + 1)^2}$
 $= \frac{2e^x}{(e^x + 1)^2}$ (2)

> 0 $\therefore f(x)$ is increasing $\forall$

(iv)

P.T.O.

(iv)



(2)

(v) slope of tangent to $y=f(x)$ at $x=0$ is $\frac{2 \cdot 0}{(0^2+1)^2} = \frac{1}{2}$

$y=mx$ will intersect $y=f(x)$ in various ways depending on whether or not $m > \frac{1}{2}$

If $m > \frac{1}{2} \Rightarrow 1 \text{ soln } (x=0)$

$0 < m < \frac{1}{2} \Rightarrow 3 \text{ solns.}$

$m < 0 \Rightarrow 1 \text{ soln } (x=0)$

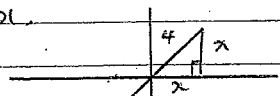
b) $xy=4 = c^2 = \frac{a^2}{2}$

$\therefore a^2 = 8$

$a = \sqrt{8}$

$c = \sqrt{2}$ (since rect. hyperbola)

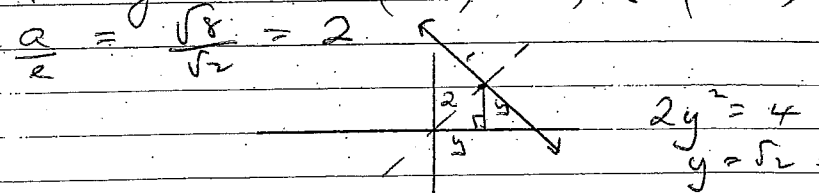
foci are $\pm ae$ units (ie 4) along $y=x$.



$2x^2 = 16$

$x = 2\sqrt{2}$

(ii) $\therefore$ foci are $(2\sqrt{2}, 2\sqrt{2})$ & $(-2\sqrt{2}, 2\sqrt{2})$



Directrix has eqn of form $x+y=K$.
 $(2, 2)$ lies on this. (1)

$K = \pm 2\sqrt{2}$

Directrices have equations $x+y = \pm 2\sqrt{2}$.

(ii) gradient chord PQ = $\frac{\frac{2}{p} - \frac{2}{q}}{2(p-q)} \times \frac{pq}{pq}$

$= \frac{2(q-p)}{2pq(p-q)}$ (2)

$= -\frac{1}{pq}$

(iv) Egn of chord PQ:

$y - \frac{2}{p} = -\frac{1}{pq}(x-2p)$ (2)

$\therefore pqy - 2q = -(x-2p)$

$\therefore x + pqy = 2(p+q)$ $\times$

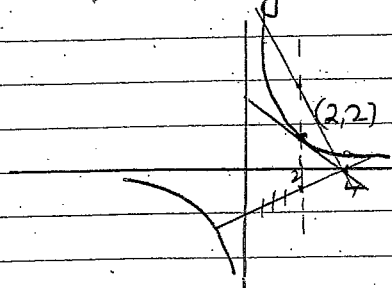
(v) $(4, 0)$ lies on $\times$

$4 = 2(p+q)$

$\therefore p+q = 2$

M: $(2, 0) + (2, 0) = (4, 0) \Rightarrow (0, 1) + (0, 1)$

(vi) ∴ locus of M is $x=2$.



(1) $y > 2$.

c) (i) $x = a \sec \theta$

$$\therefore \frac{dx}{d\theta} = a \sec \theta \tan \theta$$

$y = b \tan \theta$

$$\therefore \frac{dy}{d\theta} = b \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{b \sec^2 \theta}{a \sec \theta \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

∴ slope of normal at P is $-\frac{a \tan \theta}{b \sec \theta}$

eqn of normal is

$$y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$\left(x \frac{b}{\tan \theta} \right)$$

$$\text{i.e. } \frac{by}{\tan \theta} - b^2 = -\frac{a \sec x}{\sec \theta} + a^2$$

$$\therefore \frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 \quad (3)$$

(ii) at A, $y=0$

$$\therefore \frac{ax}{\sec \theta} = a^2 + b^2$$

$$\therefore x = \frac{a^2 + b^2}{a} \sec \theta$$

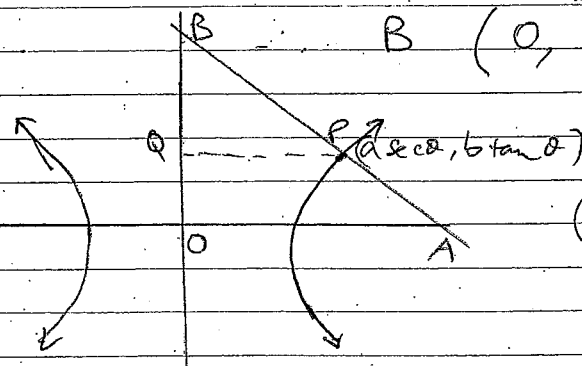
$$\text{i.e. } A \left(\frac{a^2 + b^2}{a} \sec \theta, 0 \right)$$

at B, $x=0$

$$\therefore \frac{by}{\tan \theta} = a^2 + b^2$$

$$\therefore y = \frac{a^2 + b^2}{b} \tan \theta$$

$$B \left(0, \frac{a^2 + b^2}{b} \tan \theta \right)$$



$$PA : PB = OQ : QB$$

(ratio of intercepts on transversals cut by family of parallel lines - or similar triangles argument)

$$\therefore PA : PB = b \tan \theta : \frac{a^2 + b^2}{b} \tan \theta - b \tan \theta$$

$$= b \tan \theta : \frac{a^2 + b^2}{b} \tan \theta - b \tan \theta$$

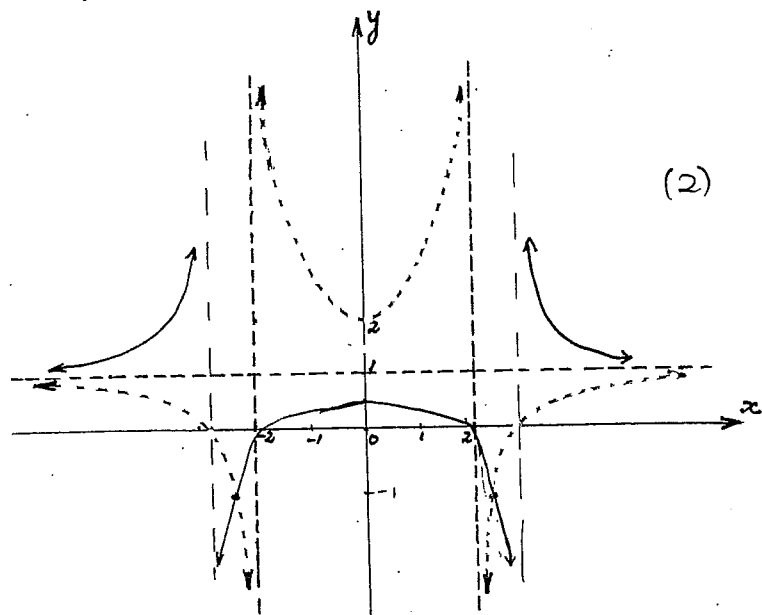
(2)

$$= b \tan \theta : \frac{a^2 \tan^2 \theta}{b}$$

$$= b^2 \tan^2 \theta : a^2 \tan^2 \theta$$

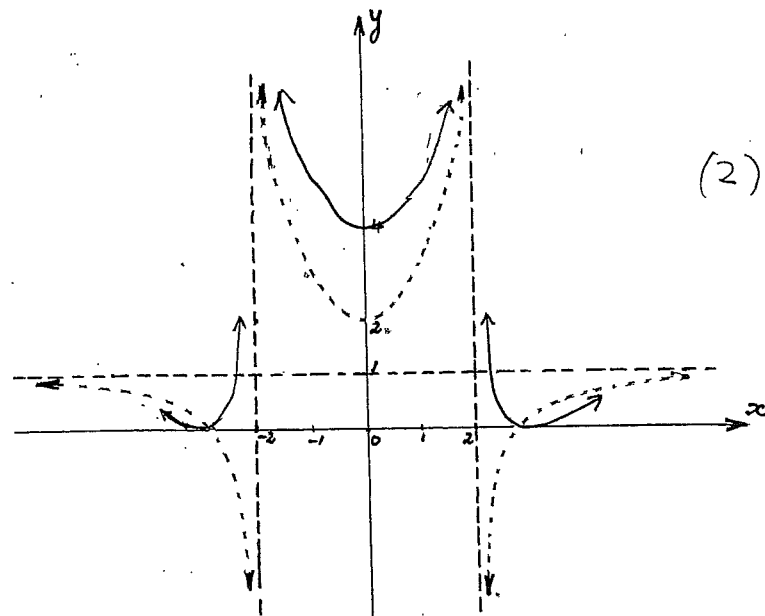
$$= b^2 : a^2$$

Q2 a) (i) $y = \frac{1}{f(x)}$



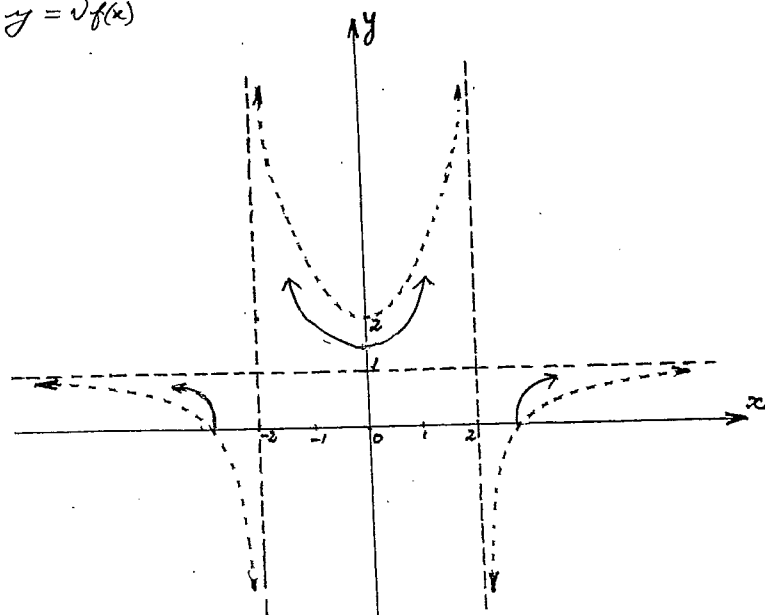
(2)

Q2 a) (iii) $y = [f(x)]^2$

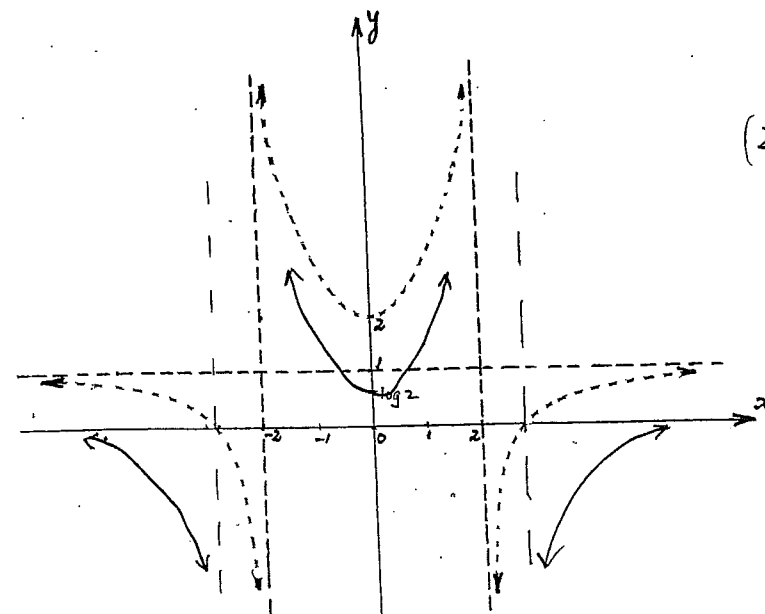


(2)

Q2 a) (ii) $y = \sqrt{f(x)}$

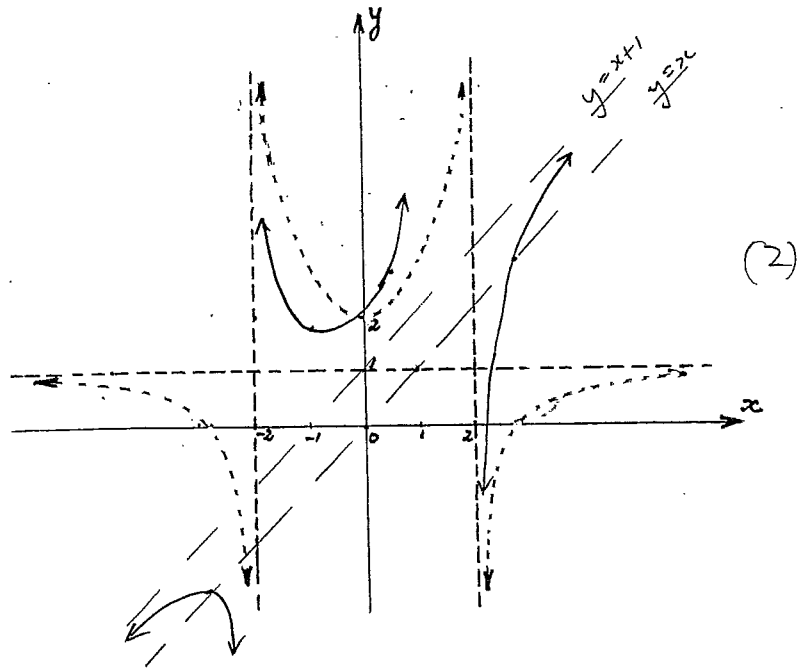


Q2 a) (iv) $y = \log[f(x)]$



(2)

Q2 a) (v) $y = x + f(x)$



(2)

Q2 a) (vi) $y = f'(x)$

